

Property-(u) ℓ_1 -preduals must contain c_0

Run `fa_banach_001`, arXiv:1504.08223, candidate partial result

Source and Question

The source paper is S. A. Argyros, I. Gasparis, and P. Motakis, *On the structure of separable $\mathcal{L}_\infty$ -spaces*, arXiv:1504.08223 [1]. In the introduction, the authors describe their asymptotic c_0 isomorphic ℓ_1 -predual without c_0 as a step toward Godefroy–Li Question IV.2:

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[LR], [JRZ], [LS], [S]) we prove that every separable infinite dimensional $\mathcal{L}_\infty$ -space is isomorphic to such a space. We use this result to deduce that every separable infinite dimensional $\mathcal{L}_\infty$ -space X has an infinite dimensional $\mathcal{L}_\infty$ -subspace Z , so that the quotient X/Z is isomorphic to c_0 .

In the final section of this paper we provide an example of an asymptotic c_0 isomorphic ℓ_1 -predual (i.e. a space whose dual is isomorphic to ℓ_1) $\mathfrak{X}_0$ that does not contain c_0 . This result in particular yields that the proximity of a Banach space to c_0 in a local setting, in the sense of being $\mathcal{L}_\infty$, as well as an asymptotic setting, in the sense of being asymptotic c_0 , does not imply proximity to c_0 in an infinite dimensional level. We think of this example as a step towards the solution of a problem in [GL] Question IV.2], namely whether every isomorphic ℓ_1 -predual satisfying Pełczyński's property-(u) is isomorphic to c_0 . The question of the existence of an asymptotic c_0 $\mathcal{L}_\infty$ -space not containing c_0 was asked by B. Sari.

2. BOURGAIN-DELBAEN SPACES

In the present section we give a general definition of the spaces that shall be called Bourgain-Delbaen spaces and prove some of their basic properties. It turns out that every infinite dimensional separable $\mathcal{L}_\infty$ -space is isomorphic to a Bourgain-Delbaen space. We recall the definition of a $\mathcal{L}_\infty$ -space, which was introduced in [LP] Definition 3.1].

The relevant source question is whether every isomorphic ℓ_1 -predual satisfying Pełczyński's property-(u) is isomorphic to c_0 .

Result

Status. This packet proves a partial obstruction, not the full Godefroy–Li classification. It shows that property-(u) forces the presence of c_0 in every infinite-dimensional isomorphic ℓ_1 -predual. The remaining question is whether the whole space must be isomorphic to c_0 .

Definitions

We use the standard formulation of Pełczyński's property-(u): for every weakly Cauchy sequence (x_n) in X , there is a weakly unconditionally Cauchy series $\sum_{k=1}^\infty u_k$ in X such that

$$x_n - \sum_{k=1}^n u_k \longrightarrow 0$$

weakly. Recall also the Bessaga–Pełczyński theorem: a Banach space contains no isomorphic copy of c_0 if and only if every weakly unconditionally Cauchy series in it is unconditionally norm convergent [2, 3].

Proof Intuition

Property-(u) says that weakly Cauchy sequences are controlled by partial sums of weakly unconditionally Cauchy series. The only obstruction to turning such partial sums into actual norm-convergent sums is c_0 : without c_0 , the Bessaga–Pełczyński theorem makes every weakly unconditionally Cauchy series unconditionally convergent. Thus property-(u) plus no c_0 upgrades weak Cauchy sequences to weakly convergent sequences.

For an ℓ_1 -predual, the dual is separable, so the space cannot contain ℓ_1 . Rosenthal’s ℓ_1 -theorem then says that any nonreflexive bounded sequence has a weakly Cauchy subsequence; weak sequential completeness would make that subsequence weakly convergent, contradicting Eberlein–Smulian. Hence no- c_0 would force reflexivity, impossible for an infinite-dimensional space whose dual is ℓ_1 .

Lemma 1. *If X has property-(u) and contains no isomorphic copy of c_0 , then X is weakly sequentially complete.*

Proof. Let (x_n) be a weakly Cauchy sequence in X . By property-(u), there is a weakly unconditionally Cauchy series $\sum u_k$ such that

$$x_n - s_n \rightarrow 0 \quad \text{weakly}, \quad s_n = \sum_{k=1}^n u_k.$$

Since X contains no c_0 , the Bessaga–Pełczyński theorem implies that $\sum u_k$ is unconditionally norm convergent. Hence $s_n \rightarrow s$ in norm for some $s \in X$. Therefore $x_n = (x_n - s_n) + s_n \rightarrow s$ weakly. $\square$

Theorem 1. *Let X be an infinite-dimensional Banach space such that X^* is isomorphic to ℓ_1 . If X has Pełczyński’s property-(u), then X contains an isomorphic copy of c_0 .*

Proof. Assume toward a contradiction that X contains no copy of c_0 . By Lemma 1, X is weakly sequentially complete.

Since $X^* \simeq \ell_1$, the dual X^* is separable. Thus X contains no copy of ℓ_1 : if $J : \ell_1 \rightarrow X$ were an isomorphic embedding, then $J^* : X^* \rightarrow \ell_\infty$ would be a bounded surjection, making ℓ_∞ a separable quotient of the separable space X^* , a contradiction.

We claim that X is reflexive. If not, the closed unit ball of X is not weakly compact. By Eberlein–Smulian there is a bounded sequence in X with no weakly convergent subsequence. Since X contains no ℓ_1 , Rosenthal’s ℓ_1 -theorem gives a weakly Cauchy subsequence. Weak sequential completeness makes this subsequence weakly convergent, a contradiction. Hence X is reflexive.

But if X is reflexive, then X^* is reflexive. This is impossible because $X^* \simeq \ell_1$ and infinite-dimensional ℓ_1 is not reflexive. Therefore the assumption that X contains no c_0 was false. $\square$

Corollary 1 (Complemented c_0 self-reduction). *Let X satisfy the hypotheses of Theorem 1. Then*

$$X \simeq c_0 \oplus Y$$

for a complemented subspace Y of X . Moreover, either Y is finite-dimensional, in which case $X \simeq c_0$, or Y is again an infinite-dimensional Banach space with property-(u) and $Y^ \simeq \ell_1$.*

Proof. By Theorem 1, X contains a subspace $G \simeq c_0$. Since $X^* \simeq \ell_1$, the space X is separable. Sobczyk’s theorem therefore gives a bounded projection $P : X \rightarrow G$. Put $Y = \ker P$. Then $X \simeq G \oplus Y \simeq c_0 \oplus Y$.

If Y is finite-dimensional, then $c_0 \oplus Y \simeq c_0$. Assume that Y is infinite-dimensional. Property-(u) passes to complemented subspaces: if (y_n) is weakly Cauchy in Y , apply property-(u) in X and then apply the projection $I - P$ to the witnessing weakly unconditionally Cauchy series. Thus Y has property-(u).

Finally, Y^* is isomorphic to a complemented subspace of $X^* \simeq \ell_1$. By Pełczyński’s theorem on complemented subspaces of ℓ_1 , every infinite-dimensional complemented subspace of ℓ_1 is isomorphic to ℓ_1 [5]. Hence $Y^* \simeq \ell_1$. $\square$

Verification and Novelty Check

The proof is purely structural and uses no computation. The cheap run indexes were searched for arXiv:1504.08223 and for the phrases “property-(u)”, “Question IV.2”, and isomorphic ℓ_1 -predual; no existing packet for this exact partial theorem was found. A nearby packet on the Rosenthal–Odell quotient problem records a broader isomorphic ℓ_1 -predual counterexample context, but not the property-(u) obstruction proved here. A bounded web/local literature check on 2026-07-03 located standard definitions and the Bessaga–Pełczyński/Rosenthal tools, but no later exact answer to the full Godefroy–Li isomorphism question.

Limitations

The theorem and Corollary 1 do not prove that X is isomorphic to c_0 . They rule out the possibility that a counterexample to Godefroy–Li Question IV.2 could avoid c_0 entirely, and they show that any remaining counterexample recursively splits off complemented copies of c_0 . The missing step is uniform control or a structural argument that forces the complemented remainders to terminate or be absorbed by c_0 .

Human Review Recommendation

Review as a likely valid partial result. The main checks are the exact definition of property-(u), the use of the Bessaga–Pełczyński characterization of spaces with no c_0 , and the separable-dual exclusion of ℓ_1 subspaces.

References

- [1] S. A. Argyros, I. Gasparis, and P. Motakis, *On the structure of separable $\mathcal{L}_\infty$ -spaces*, arXiv:1504.08223, 2015.
- [2] F. Albiac and N. J. Kalton, *Topics in Banach Space Theory*, Graduate Texts in Mathematics 233, Springer, 2006.
- [3] C. Bessaga and A. Pełczyński, *On bases and unconditional convergence of series in Banach spaces*, Studia Math. **17** (1958), 151–164.
- [4] H. P. Rosenthal, *A characterization of Banach spaces containing ℓ_1* , Proc. Nat. Acad. Sci. U.S.A. **71** (1974), 2411–2413.

- [5] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces I: Sequence Spaces*, Ergebnisse der Mathematik und ihrer Grenzgebiete 92, Springer, 1977.